Engineering Chemistry · Unit_03 · Energy storage devices, hydrogen energy, and sensors

DRAFT — UE26CY151A · Unit_03 · Energy storage devices, hydrogen energy, and sensors · Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2))

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
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M1

Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2))

1 Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2))

Study focus: Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000150 · EV0000151

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module 4 on electrochemical equilibria.

Key details

- We look at the construction, the working and how to use the glass electrode to determine pH of a solution.
- It's an ion selective electrode which responds to H plus ions.
- And therefore we can say that it is pH sensitive.
- So like any other ion selective electrode, there has to be a membrane which is going to be able to exchange these H plus ions in this case because we want to determine pH of the solution and here it is the glass membrane.
- We have a bulb.
- Now, this bulb is the most important part of the glass electrode because this is made of a very thin membrane of glass, and this glass is of a specific type, and that is corning 015.
- It is used to make this glass.
- The bulb the glass bulb is made up of coning 015, and this is a very thin membrane.
- We need a reference solution.
- And that is.
- HCl of 0.1 molar concentration.
- And in and in this solution, there is the internal reference electrode that is the silver chloride.

- And then we have the internal reference solution of HCl of 0.1 molar concentration, and then we have the internal reference electrode, which is the silver chloride electrode.
- We have to place this glass electrode in a solution which contains H plus, the concentration of which is not known to us, and therefore using this glass electrode we are going to find out.
- the concentration of the H plus ion in the analy solution.
- If you write only this, then that would mean that we are talking about the silver chloride electrode.
- But since glass electrode and we know it is only the glass which is capable of exchanging these H plus ions and not the silver chloride electrode.
- Therefore, we have to mention the glass here.
- That is the electrode representation.
- This is the schematic representation, just like we have for the ion selective electrode, we have the membrane here, and on one side of the membrane we have the reference solution.
- In this case, it is 0.1 molar HCl.
- That is H plus solution plus Na plus glass.

Example or application from the lecture

- And in today's class we are going to study the glass electrode, which is an example for ion selective electrode.
- Let us look at the construction of this glass electrode.
- Let us look at the working of this glass electrode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 05:00–09:59

If you look at the structure happens to be Na plus Gl minus, and that is in equilibrium with Na plus solution.

Key details

- Plus h plus gl minus.
- What can you understand from this reaction?
- That NA plus Gl minus, which is glass, is being able to exchange this H plus.
- Plus goes into the solution.
- This is the exchange reaction which is taking place at the membrane, at the surface of the why is this possible?
- If you look at the structure of glass, stru glass is actually a three-dimensional network of silica tetra hetra.
- And since an open network structure we have this interstitial space which are filled with cations.
- It is going to depend on how the glasses.
- Is made if it's sodium silicate, we are going to have sodium ice.
- If it's alumino silicate, we can have aluminium ice, calcium silicate, calcium ice.
- If it's alumino, potassium silicate, we'll have both aluminum as well as K plus ice.

- This membrane comes in contact with the solution containing H plus ions.
- What happens is.
- The outer lay of this glass membrane are going to swell up.
- When they absorb this solution and they swell up.
- These H plus ions will go in and the sodium ions will come out.
- But mostly the divalent and the trivalent are held very tight.
- So therefore, it's only the monovalent ions which are found to be mobile.
- The H plus ion now you can see in the picture over here, since it is acidic medium, we find that H plus ion moves into this gel layer and the NA plus ion is going to come into the solution.
- What do we mean by exchange of these ions?
- From the solution, and based on the concentration of H plus ion, a potential is going to be developed at the membrane.
- At the membrane, we have a potential which is getting formed.
- Get an expression for the potential of the glass electrode.
- where we know it is 0.1 molar HCl.
- Again, this is internal reference electrode that is AG ADCl electrode.
- On the other side of the membrane, we have the analyte solution where we have H plus concentration is equal to C1.
- surface and this membrane surface are in contact with different concentration of H plus ions.
- The concentration C2, the outer membrane is in contact with the concentration C1.
- Now, because of this, at the outer membrane we get a potential E1, at the inner membrane we get a potential E2.
- At the boundary now, therefore, at the membrane boundary, we have a difference in the potential that is E1 minus E2.
- Because the membrane.
- However thin it is, it has an inner surface and it has an outer surface.
- The inner surface is in contact with one concentration of H plus, the outer surface is in contact with a different concentration of H plus.

Example or application from the lecture

- First let us understand that.

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Core explanation · 10:00–14:59

Now C2 is a constant because it is an internal reference solution.

Key details

- We already know it is 0.1 molar H Cl.
- Say the constant is L dash plus 2.303 RT over N F log of C1.
- It is minus of 2.303 R T over N F log of C2.
- nine one over n log of h plus now since.

- For H plus I n is equal to 1.
- Your expression now for the boundary potential we got L dash minus 0.0591 p h.
- We had the boundary potential plus the potential due to the internal reference electrode, which has been placed there to provide the electrical contact.
- Apart from this boundary potential and the internal reference electrode potential, we have a third potential coming up that is known as the asymmetric potential.
- So, therefore, the sum of boundary potential, reference electrode potential, internal reference electrode potential, and the asymmetric potential.
- Potential will give us the total glass electrode potential.
- We know that the glass bulb is curved, so the radius of curvature itself is different for the inner and the outer membrane.
- So what happens because of that?
- There is a small potential.
- Which is known as asymmetry potential.
- outside the analyte solution as the same concentration as the reference solution, we should not be able to measure any other potential except the potential due to the internal reference electrode.
- But we find that we even though both the concentrations are same, there exists a very small potential which is known.
- As the asymmetric potential, and that is due to the membrane itself, and it is.
- We get this expression.
- are all constants.
- So here we have E0 G is equal to L dash plus E reference plus the E asymmetric, and therefore we get the final expression for the glass electrode that is EG is equal to e naught g minus 0.0591 ph so if I know if I can know what is EG, E0 G being constant, I can.
- Find out what is the pH of the solution.
- But since this is a single electrode, we know that single electrode potentials cannot be measured.
- Therefore, we need to make a combination, and now what we have is we have to combine it with the external reference solution.
- solution using a solid bridge.
- You can see here this is the calomel electrode, which is the external reference electrode, and this is the glass electrode.

Example or application from the lecture

- Let us understand this.

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EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 15:00–19:59

cell representation for this combination.

Key details

- And why are we doing this?
- Because we want to see how to use the glass electrode to determine pH of the solution.
- We have the.
- External reference electrode that is the calomide electrode HE2Cl2 Cl minus.
- Through the solid here, through the porous grid here, it is in contact with the analy solution.
- That in turn is in contact with the solution, that is the internal reference solution, and that is in contact with the silver superchloride electrode.
- We have galomel electrode which is diped in analyte that is in contact with the glass which is in contact with 0.1 normal HCl or 0.1 molar HCl and that is in contact with the silver chloride and then silver.
- Using the glass electrode, how are we determining the pH?
- We know that we have made a combination and we can measure the EMF of this cell.
- This can be measured potentiometrically.
- E right minus E left, so E glass minus E calomel.
- E glass just now we got this expression that it is E naught G minus 0.0591 PH and when we substitute we get this expression E naught G minus 0.0591 PH minus E calomel and if you rearrange we get the expression for pH.
- E naught G is fixed for a particular glass electrode and then using this expression I can determine the pH of the solution.
- All the glass electrodes.
- I mean what I mean is E0 G is not the same for all the glass electrode.
- Therefore, before using any glass electrode, I need to measure what is E0 G for that particular glass electrode.
- Using the same expression or the same setup, I can do it.
- Do I do now?
- I take a solution of known pH.
- If I know pH and I know what is e calomel, I can measure E cell, I can determine what is E0 G.
- So this is how we are going to determine E0 G first, and that can be used for the to determine the.
- further once we know what is E0 G, then we can determine pH of any solution.
- That is with the help of a buffer solution.
- We use a solution of known pH, usually a constant pH like buffer solution, and we find out E0 G and.
- And further, knowing ERG we can just measure the E cell and get the values of the various solution the hydrogen and concentration of different solution.
- It is used extensively in chemical, industrial, agricultural, and biological labs.
- What are the advantages?
- It can be used in oxidizing and reducing environments.
- Which was not possible with the help of hydrogen electrode.
- Hydrogen electrode can also be used to determine H plus, but it cannot be used in oxidizing and reducing environment environments.
- And metal ions, of course, we know it's an ion selective electrode.
- It is going to.

- be sensitive or detect only the H plus ions and not the other metal ions, and it does not get poison unlike the hydrogen electrode, and it can be used for very small volumes.
- And one of the most important.
- Advantage is that we get accurate results up to a pH of nine.
- However, beyond the pH of 9, we need special type of glass electrodes which can measure up to 14 pH with a good amount of accuracy.
- And they are of course very simple to operate and can be used with portable instruments.
- What are the disadvantages of glass electrode?
- And whenever there is a membrane, it offers very high resistance.
- The same thing happens with the glass electrode also.
- It is common for any ion selective electrode.
- potentiometers for EMF measurements and secondly the glass membrane which we are using is a very thin membrane so it is very delicate and hence has to be handled very carefully.

Example or application from the lecture

- Let us find an expression now.

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Core explanation · 20:00–24:59

Just now I mentioned up to pH of 9 it gives very good accurate results, but beyond the pH of 9, we see some error in the normal in the glass electrode, and that electric is referred to as the alkaline error.

Key details

- reaction which normally happens at in when there is a glass electrode in contact with h plus ions.
- And if sodium ions are more in the solution, then the reverse reaction happens.
- It's an equilibrium.
- Going into that means the electrode becomes responsive to Na plus ions.
- Therefore, the pH now which you measure is going to be lower than the.
- So what you measure will be lower because now it is responding to NA plus ions also.
- So that is about the glass electrode and in the next class we are going to look at another ion selective electrode that is the fluoride ion selective electrode.

Example or application from the lecture

- The NA plus ion gets into the into the glass and the H plus ion is now given out.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00034	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00035	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00036	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00037	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

C Revision and retrieval practice